

Test pyPDFCodebook with Images

Table of contents:

Project Overview: Summary of Project Details	3
Data Dictionary: Summary of Variables	4
Variable Details and Notes	5
Key Terms and Definitions	11

Test pyPDFCodebook with Images

Test pyPDFCodebook with Images

Project Overview: Summary of Project Details

Project Title: PDF Codebook

Principal Investigator: Nathanael Rosenheim: nrosenheim@arch.tamu.edu

Required Citations:

Nathanael Rosenheim. (2025). npr99/pypdfcodebook: v0.5.1 Zenodo Release (v0.5.1). Zenodo. <https://doi.org/10.5281/zenodo.17956283>

Project Description : pyPDFCodeBook helps researchers and data professionals create clear, attractive codebooks for tabular datasets. Codebooks document essential metadata—project descriptions, data provenance, variable definitions, and summaries—ensuring your data is understandable and reproducible. While datasets contain values, they rarely explain what each column or row represents. Codebooks fill this gap by providing structured, self-explanatory documentation. They reinforce best practices such as tidy data, unique keys, and transparent variable origins. With pyPDFCodeBook, generating a professional, easy-to-read codebook takes just a few clicks—laying the foundation for good data science and reproducible research. Remember: You are your number one data user, so help your future self out and document your metadata.

Funding Source: Award name and number:

Center for Risk-Based Community Resilience Planning : NIST-70NANB20H008, NIST-70NANB15H044 (<http://resilience.colostate.edu/>)

Southeast Texas Urban Integrated Field Lab: DE-SC0023216 (<https://setx-uifl.org/>)

Related Works:

Keywords: Codebook, data dictionary, metadata

License: Open Data Commons Attribution License (<https://opendatacommons.org/licenses/by/summary/>)

Test pyPDFCodebook with Images

Data Dictionary: Summary of Variables

Variable Name	Data Type	Length	Categorical	Variable Label
huid	String			Housing Unit ID
blockid	String	15		Block ID
numprec	Int			Number of Person Records
ownership	Int		True	Tenure Status

Test pyPDFCodebook with Images

Variable Details and Notes

The following pages provide details on each variable. Where applicable, notes provide links to verify data. Categorical variables include details on category codes.

Test pyPDFCodebook with Images

huid: Housing Unit ID

Variable characteristic	Variable details
variable type	string
total cases	49
valid cases	49
missing cases	0
unit of measure	Housing units
unit of analysis	Housing unit
unique values	49
minimum length	20
maximum length	20
example 1	B371559601011027H001
example 2	B371559601011013H001
example 3	B371559601011031H005
example 4	B371559601011031H051

Variable Notes: huid

1. Primary Key for sample data.

Test pyPDFCodebook with Images

blockid: Block ID

Variable characteristic	Variable details
variable type	string
total cases	49
valid cases	49
missing cases	0
unit of measure	Housing unit in census block
unit of analysis	Geographic unit
unique values	1
minimum length	17
maximum length	17
example 1	371560000000000.0
example 2	371560000000000.0
example 3	371560000000000.0
example 4	371560000000000.0

Variable Notes: blockid

1. 2010 Census Block ID

Test pyPDFCodebook with Images

numprec: Number of Person Records

Variable characteristic	Variable details
variable type	numeric (Int)
total cases	49
valid cases	49
missing cases	0
unit of measure	Persons
unit of analysis	Housing unit
range	minimum value: 1.00 to maximum value: 1.00
mean	1.00
median	1.00
standard deviation	0.00
10th percentile	1.00
25th percentile	1.00
50th percentile	1.00
75th percentile	1.00
90th percentile	1.00

Test pyPDFCodebook with Images

ownership: Tenure Status

Variable characteristic	Variable details
variable type	categorical (Int)
total cases	49
valid cases	49.0
missing cases	0
unit of measure	Housing unit
unit of analysis	Household
range	minimum value: 1 to maximum value: 2

ownership: Tenure Status - Categorical codes, labels and frequencies

Code	Label	Count of Housing units	Percent Housing units	Sum of Persons	Percent Persons
1	1. Owned or being bought (loan)	33	67.35%	33	67.35%
2	2. Rented	16	32.65%	16	32.65%

Variable Notes: ownership

1. Based on 2010 Census SF1 Table H16.
2. Tenure status is not applicable for vacant not occupied housing units.
3. Tenure status is not applicable for group quarters.
4. To verify results compare table to:
<https://data.census.gov/cedsci/table?q=0500000US37156&tid=DECENNIALSF12010.H16>.

Test pyPDFCodebook with Images

Test pyPDFCodebook with Images

Key Terms and Definitions

Codebook: "A codebook describes the contents, structure, and layout of a data collection... The main body of a codebook contains unambiguous variable level details. " (ICPSR: <https://www.icpsr.umich.edu/sites/icpsr/posts/shared/what-is-a-codebook>).

Data Dictionaries: A compact list of the variables in the dataset. Includes variable name, data type, length, if the variable is categorical or not, and the variable label.

Key: "A key is a variable or set of variables that uniquely identifies the elements of a table. The variables that form the key never take on missing values, and a key's value is never duplicated across rows of the table." (Gentzko & Shapiro (2014). Code and Data for the Social Sciences: A Practitioner's Guide. p. 20 <https://web.stanford.edu/~gentzkow/research/CodeAndData.pdf>)

Metadata: "Metadata is information about the data, descriptors that facilitate cataloguing data and data discovery." (The Turing Way. "Metadata in Research Data Management." The Turing Way Book: <https://book.the-turing-way.org/reproducible-research/rdm/rdm-metadata>).

Provenance: "The precise subset and version of data a set of result originates from. (The Turing Way Book: <https://book.the-turing-way.org/reproducible-research/vcs/vcs-data/>).

Tabular Dataset: "Datasets made up of rows and columns." (Tidy data: <https://tidyr.tidyverse.org/articles/tidy-data.html>)

Tidy Data: "Each variable is a column; each column is a variable. Each observation is a row; each row is an observation. Each value is a cell; each cell is a single value." (Tidy data: <https://tidyr.tidyverse.org/articles/tidy-data.html>)

Unit of Observation: "The object(s) of analysis for the data collection, such as an organization, individual, or household." (ICPSR: https://icpsr.github.io/metadata/icpsr_study_schema/#unit_of_observation).

Test pyPDFCodebook with Images